

Compile hourly LWC data

1. Compile all hourly LWC data for 2009 through 2017 from ALSC (e.g. [WFC.2009.MET.R.xlsx](#))
2. Adjust minute resolution ASRC LWC data (2018 through 2025) for baseline drift, based on elevated values during clear air periods (see [Adjustments](#), below)
3. Calculate hourly avg ASRC LWC values (2018 through 2025, not previously published elsewhere)
4. Filter out LWC values below zero or above 2.0 g/m³ from the hourly average datasets (set to NaN)
5. Time sync and evaluate hourly avg LWC data for 2022 through 2025 based on PVM/visibility sensor comparison
6. Fill gaps in Gerber PVM dataset with LWC calculated based on visibility sensor readings (see [Merge_LWC_Vis](#), below)
7. Append ASRC hourly average merged LWC data to ALSC data

Calculate in-cloud LWC average values associated with cloud water samples

1. Compile list of DumpDateTimes and SampleIDs for all cloud water samples

- 1.1 Start with Chris GitHub dateW and LabID data for 2009 through 2017 ([AllCloudandMetData.csv](#))

dateW = DateON (start time for cloud water sampling interval)

convert dateW to Dump_DateTime (end time for cloud water sampling interval)

for 2009 through 2013, Dump_DateTime = DateON + 3 hrs

for 2014 through 2017, Dump_DateTime = DateON + 12 hrs

dateW in 2014 are incorrect (neither DumpDateTime NOR DateON)

DumpDateTime in 2014 now calculated based on hourly Pooled Volume dataset (see [DumpTimes2014](#), below)

- 1.2 Append Archana GitHub DumpDateTime and ID data for 2022 through 2024 (e.g. [2018.xlsx](#))

- 1.3 Add 2025 Dump_DateTime and Sample IDs from 2025 cloud water master list

2. Calculate average in-cloud LWC average for each cloud water sample (see [calc_avg_incloud_LWC2](#), below)

3. Perform linear fit of in-cloud LWC averages

4. Calculate annual means for in-cloud LWC averages each summer (see [calc_annual_mean_sampleLWC](#), below)

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Function calc_avg_incloud_LWC2()
    wave LWC = $"LWC_filt" // in LWC_hrly_ALSC-ASRC folder (hourly data)
    wave DateTimeW = $"Date_Time"
    variable NumMetHrs = DimSize(LWC,0)
    wave Dump_DateTime
    variable Samples = DimSize(Dump_DateTime,0)

    make/O/N=(Samples) LWC_avg = 0
    make/O/N=(Samples) NumHrs = 0
    make/O/N=(Samples) YearW = 0

    variable row, sam
    for (sam = 0; sam < Samples; sam += 1)
        String dateStr = Secs2Date(Dump_DateTime[sam], -2) // Returns "YYYY-MM-DD"
        Variable year, month, day
        sscanf dateStr, "%d-%d-%d", year, month, day
        YearW[sam] = year

        for (row = 0; row < NumMetHrs; row += 1)
            if ((YearW[sam] > 2008) && (YearW[sam] < 2014)) // for samples 2009 through 2014
                if ((DateTimeW[row] >= Dump_DateTime[sam]-(3*60*60)) && (DateTimeW[row] < Dump_DateTime[sam]+(0*60*60)))
                    if ((LWC[row] > 0.05))
                        LWC_avg[sam] += LWC[row]
                        NumHrs[sam] += 1
                    endif
                endif
            elseif (YearW[sam] > 2013) // for samples 2014 through 2025
                if ((DateTimeW[row] >= Dump_DateTime[sam]-(12*60*60)) && (DateTimeW[row] < Dump_DateTime[sam]-(0*60*60)))
                    if ((LWC[row] > 0.05))
                        LWC_avg[sam] += LWC[row]
                        NumHrs[sam] += 1
                    endif
                endif
            endif
        endfor
        LWC_avg[sam] /= NumHrs[sam]
    endfor
End
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Function DumpTimes2014() // determine actual dump times based on hourly cloud water data
    wave POOL_VOL_m1, COL_DATE_TIME, IND_VOL_m1
    wave/T LABNO
    variable NumRows = DimSize(POOL_VOL_m1,0)
    make/D/O/N=(NumRows) DateON2014 = 0
    make/D/O/N=(NumRows) SampleVol2014 = 0
    make/T/O/N=(NumRows) SampleID2014
    variable row, sam = 1

    DateON2014[0] = COL_DATE_TIME[0]
    SampleVol2014[0] = POOL_VOL_m1[0]
    SampleID2014[0] = LABNO[0]

    for (row = 0; row < NumRows; row += 1)
        if((POOL_VOL_m1[row] == 250))
            DeletePoints row,1, COL_DATE_TIME,IND_VOL_m1,POOL_VOL_m1,LABNO
            row -= 1
            NumRows -= 1
        endif
    endfor

    for (row = 1; row < NumRows; row += 1)
        if (POOL_VOL_m1[row] == POOL_VOL_m1[row-1])
        else
            DATEON2014[sam] = COL_DATE_TIME[row]
            SampleVol2014[sam] = POOL_VOL_m1[row]
            SampleID2014[sam] = LABNO[row]
            sam += 1
        endif
    endfor
End
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Function Merge_LWC_Vis() // for each year separately 2022 through 2025 (from GCVI folder in each year)
    wave LWC_avg, start_3600sec
    wave vis_recv_exp
    variable NumRows = DimSize(start_3600sec,0)
    variable row = 0

    make/O/N=(NumRows) LWC_calc = 0
    make/O/N=(NumRows) Merged_LWC = LWC_avg

    for (row = 0; row < NumRows; row += 1)
        LWC_calc[row] = (vis_recv_exp[row]-0.475)/22.5 // based on 2022 calibration

        if ((NumType(LWC_avg[row]) == 2) || (LWC_avg[row] < 0))
            Merged_LWC[row] = LWC_calc[row]
        endif
    endfor

    Merged_LWC = Merged_LWC < 0 ? 0 : Merged_LWC
End
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Function calc_annual_mean_sampleLWC() // for use with full longterm dataset (LWC only)
    wave LWC_avg, YearW
    variable NumRows = DimSize(LWC_avg, 0)

    variable NumYears = 2025-2009 + 1
    make/O/D/N=(NumYears) LWC_annual_mean = 0
    make/O/D/N=(NumYears) numLWCpts = 0

    variable row, yr
    for (row = 0; row < NumRows; row += 1)
        for (yr = 0; yr < NumYears; yr += 1)
            if (YearW[row] == yr + 2009)
                if (NumType(LWC_avg[row])>0)
                    LWC_annual_mean[yr] += LWC_avg[row]
                    numLWCpts[yr] += 1
                endif
            endif
        endfor
    endfor

    LWC_annual_mean /= numLWCpts

    make/O/D/N=(NumYears) LWC_annual_stdev = 0

    for (row = 0; row < NumRows; row += 1)
        for (yr = 0; yr < NumYears; yr += 1)
            if (YearW[row] == yr + 2009)
                if (LWC_avg[row]>0)
                    LWC_annual_stdev[yr] += (LWC_avg[row]-LWC_annual_mean[yr])^2
                    numLWCpts[yr] += 1
                endif
            endif
        endfor
    endfor
    LWC_annual_stdev = sqrt(LWC_annual_stdev/numLWCpts)
End
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Adjustments

1-min resolution LWC data in 2018 was shifted +0.04 Aug 3 - 8
 1-min resolution LWC data in 2018 was shifted -0.04 Aug 10-Aug 17
 1-min resolution LWC data in 2020 was shifted -1.63 July 15-16
 1-min resolution LWC data in 2020 was shifted -0.08 June 23-25
 1-min resolution LWC data in 2022 was shifted +0.1 June 19 - July 11
 1-min resolution LWC data in 2023 was shifted -0.02 June 20 - Sept 29
 1-min resolution LWC data in 2023 was shifted +0.01 July 6-19
 1-min resolution LWC data in 2024 was shifted -0.01 June 14-17
 1-min resolution LWC data in 2024 was shifted -0.02 June 17-22
 1-min resolution LWC data in 2024 was shifted -0.058 June 26-28

Then, 1-min resolution LWC data averaged hourly in 2018-2025

- AveragXSecs("DateTimeW","LWC_value","LWC_hrlyavg","LWC_hrlystd","LWC_hrlynum",3600)